

Beyond the Sharing of Personal Data: Design of Collaborative Chatbot-based Mobile Support Systems to Facilitate Recovery from Alcohol Use Disorder

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Abstract

Alcohol Use Disorder (AUD) is a chronic condition with profound negative consequences for one's health and personal relationships. In this pilot needs-finding study, we interviewed four AUD patient-family dyads to investigate the integration of chatbot technology with existing self-monitoring and social sensing tools in AUD treatment. Insights from these interviews informed the design of mobile applications aimed at promoting collaboration between patients and family companions. A chatbot-powered backend supports self-monitoring, facilitates family updates, and provides continuous guidance via a 24/7 conversational agent. Subsequent research will involve a field study to evaluate the feasibility of the system for clinical application in AUD management.

CCS Concepts

• **Human-centered computing** → **Empirical studies in HCI**;
Empirical studies in ubiquitous and mobile computing.

Keywords

Alcohol Use Disorder, Social Sensing, Chatbot Technology.

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1 Introduction

Alcohol Use Disorder (AUD) refers to the compulsive consumption of alcohol, affecting one's health, professional responsibilities, and interpersonal relationships [15]. The recovery process typically involves inpatient detoxification followed by outpatient programs aimed at maintaining sobriety [22]. However, low treatment uptake and the limited frequency and scope of outpatient interactions hinder long-term success [14], such that nearly two-thirds of individuals relapse within six months [17].

Recent advances in natural language processing (NLP) have enabled chatbot interactions that closely resemble human conversation [5], and the integration of chatbots with mobile applications has transformed the ways that users access information [10]. Chatbots have been implemented in mental health care scenarios to provide guidance in dealing with stress [11], depression [9], and dementia [20]. This study explored the use of chatbot technology as a convenient real-time support mechanism to monitor drinking patterns [1], assess the risk of relapse [21], and provide personalized emotional support [1]. By sustaining user engagement, such tools may enhance the long-term effectiveness of AUD interventions.

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A pilot needs-finding study involving four AUD patient-family dyads identified three key themes critical to the overall design of the system. The proposed system includes a patient app for self-monitoring, a companion app for sharing progress reports, and a chatbot-powered agent providing 24/7 support. The system was designed to ensure long-term sobriety by promoting accountability and improving communication, based on the theory that family companions with access to shared data and chatbot-generated summaries are better equipped to assess situations objectively and offer meaningful support. This work is currently undergoing refinement in collaboration with psychiatrists in preparation for a field study to evaluate its feasibility and effectiveness.

2 Needs Finding Study

Needs analysis was based on one-on-one interviews with dyads comprising an AUD patient and a family companion. Four patients (referred to as S1–S4), including two women and two men with a mean age of 45.75 years (range = 33–54 years; SD = 9.11) were recruited from Taipei City Hospital. Family companions (referred to as T1–T4) were also recruited. The mean age of the companions was 53.75 (43–61 years; SD = 7.63), including the mother of S1, the older sister of S2, the wife of S3, and the husband of S4. All participants, who had completed withdrawal treatment and returned to daily life, were referred by psychiatrists who confirmed their suitability for the study.

The survey involved a technology probe session followed by semi-structured interviews. A comprehensive review of the literature [1, 6–8, 13, 18, 19, 23] revealed several ways that chatbots could be used, such as querying symptoms [18], delivering diagnosis-related information [18], and sending reminders to take medication. Chatbots can assist in setting recovery goals, tracking progress, and sharing updates with companions [19]. Chatbots can also promote patient engagement by sending empathetic messages, generating motivational content, and initiating proactive check-ins, aligned with the principles of Cognitive Behavioral Therapy (CBT) [19]. Finally, chatbots can mediate interactions and foster collaborative recovery efforts [6, 23]. They can also be personalized in terms of tone and demeanor to improve communication [2].

In accordance with these insights, we designed four chatbot usage scenarios applicable to mobile devices: (1) Monitoring the emotional state of the patient and recommending corresponding courses of action; (2) Tracking health indicators to enhance self-awareness, promote healthy habits, and prevent declines; (3) Providing companionship and emotional support during moments of loneliness or despair; and (4) Facilitating interactions with significant others. The proposed chatbot applications were simulated using ChatGPT running on a laptop, the personality of which was pre-configured by a researcher through the issuance of specific prompts. The participants interacted with the chatbot by discussing recovery-related issues (e.g., goal setting) with the aim of providing participants with a tangible sense of how such technologies could be implemented in the future.

Semi-structured interviews were also conducted with patients and family companions to identify challenges in dealing with AUD and gain insights into their experiences during the technology probe session as well as the potential opportunities provided by

chatbot technology. The interviews with patients and companions were conducted separately to encourage open discussion. Patient interviews focused on (1) their perceptions of AUD; (2) their experience in managing alcohol use; and (3) their experience engaging with family companions on the topic of AUD. Family companion interviews focused on (1) their perceptions of AUD; (2) their experience communicating with the patient about alcohol use; and (3) the methods they used to moderate patient behavior. Each interview lasted between 90 and 120 minutes.

The interviews were audio-recorded after obtaining IRB approval and participant consent. The recordings were transcribed to facilitate bottom-up thematic analysis aimed at identifying prominent themes [4], which were then connected via axial coding. Multiple rounds of open coding revealed key issues affecting communication between patients and companions under conventional AUD treatment practices, as well as ways that chatbots could be used to address these challenges.

2.1 Identifying Actionable Steps through Casual Conversations with a Chatbot

Goal setting is a crucial step in dealing with any kind of addiction. The inability of many patients to break their goals down into actionable steps can lead to an over-reliance on their doctor and limit their focus to the completion of basic tasks. S3 commented *“Frankly speaking, I never made a clear or concrete goal for quitting alcohol. I simply followed my psychiatrist’s advice and took my medication as directed.”* Goal setting is even more difficult in situations where family companions did not live with the patient. Some family companions expressed a willingness to assist in setting goals; however, most of these suggestions were based on personal observations, due to their inability to fathom the difficulties faced by patients. This asymmetry can lead to misunderstandings. T3 made the following comment: *“I initially thought about using a table or chart to make plans (laughs), but he dislikes making plans and seldom sticks to them. A chatbot might be able to get him to exercise and adopt a healthier lifestyle. But he needs clear instructions, such as exercising on Mondays and Wednesdays and learning something new on Tuesdays and Thursdays. But, without any way to monitor his status, I could not be sure about his progress. I ultimately decided to follow the doctor’s advice and let him figure things out for himself.”*

Patients expressed a need for someone with whom to discuss goal setting and the challenges of quitting alcohol. They made it clear that these conversations could not be held with their family companions, but expressed enthusiasm for engaging with a chatbot. S3 made the following comment: *“Talking to the chatbot helped to relieve my frustration — at least to some extent. The responses made sense to me and the suggestions seemed practical. At the very least, this process gave me a practical outlet for my negative emotions. So, I think chatbots could also be used to uncover negative thoughts.”* When given practical reminders by the chatbot, patients were able to rediscover meaningful activities, alleviate feelings of loneliness, and shift their focus away from negative emotions. S2 made the following comment: *“My discussion with the chatbot about goals was really helpful. When I was unsure of what to do, the chatbot provided ideas for activities, such as running or exercising. It also suggested*

things I hadn't considered, such as Jigsaw puzzles. I remembered that I still had a few puzzles at home."

2.2 Serving as an Empathetic Bridge Between Patients and Companions

Some patients expressed concern about burdening family companions with their difficulties, preferring to manage their problems on their own. S2 made the following comment: *"Depending on others for support doesn't feel right, especially since we don't live together—I live alone now. Having someone remind me regularly could be helpful, but my second sister couldn't do that consistently — she has her own life and responsibilities."* Even when sober, the patients lacked the motivation to share their physical and psychological state. Family companions reported that when they made inquiries, patients tended to respond with an uncooperative tone or avoid the discussion entirely. T4 made the following comment: *"She never mentions health issues and avoids discussing her emotions, even when she's sober. She doesn't share her inner thoughts, leaving me unaware of her motivations. I've asked her, but she doesn't provide a clear answer. If I could be informed of her physical and psychological state through other means, it would help me to understand her emotional state."*

The warmth and empathy expressed by the chatbot during these brief interactions came as a surprise to some family companions. These interactions brought out a better side of the patient, suggesting that patients would be more receptive to suggestions from a chatbot than from a close family companion. T1 made the following comment: *"When I am interacting with her, I can observe her emotions directly, and it gets me all worked up. However, the communication style of the chatbot is different. This might make it easier for her to open up and share. The consistently warm and gentle tone of the chatbot might make her more willing to communicate."* Family companions expressed frustration in communicating with patients and the compounding of negative feelings with each failed interaction. Again, the chatbot provided an opportunity to vent their emotions, offering a calming and supportive presence. T4 made the following comment: *"I often feel powerless when trying to interact with her [S4] and don't know who I can turn to. Generally, I share my worries and frustrations with my sisters, but I feel like I'm burdening them with my problems, which only adds to my frustration. However, when I interacted with the chatbot, it felt like talking to a real person—almost like a counselor guiding me step by step. This had a noticeable effect on my mood. I found it really helpful."*

2.3 Summarizing the Context of Shared Data in a Thoughtful and Comprehensive Manner

Several patients expressed concern about the inability of family companions to recognize their efforts to maintain sobriety. Viewing the patient as a stereotypical drunk can severely impede communication. S4 made the following comment: *"When talking face-to-face, my husband's [T4's] tone quickly becomes defensive. For example, when I tell him that I'm already trying so hard, he often responds with, 'How hard are you trying? Look how early you started drinking.' He disregards what I am saying, doubting that I am making an effort. Exchanges like this are a vicious cycle."* Negative face-to-face communication patterns can make patients defensive, particularly

in situations involving the assignment of blame. Under these conditions, conversations devolve into superficial exchanges that lack any meaningful discussion of inner motivations or challenges. T4 made the following comment: *"She [S4] doesn't have a job, so she doesn't have work-related pressure. She doesn't seem to have any mental health issues either. I can't figure out what caused her to start drinking. Even now, I still don't know the reason. She has never shared it with me — and when I ask, she just keeps silent."*

As noted earlier, patients were optimistic about using chatbots to perform crucial communication tasks; however, they expressed concern about the sharing of personal data (e.g., alcohol test results). Many feared that their family companions would scrutinize every message looking for hidden patterns, and that sharing incomplete information could lead to negative assumptions or imagined scenarios. S2 made the following comment: *"I am often reluctant to share information with my sister [T2], especially when I have been drinking. I need to talk to her beforehand and establish clear boundaries, such as: 'There is no need to respond to every message I send. Just quietly acknowledge what I say and move on.' I'm worried that the shared data could get her all worked up."* Patients appreciated the warm tone of the chatbot and its ability to provide easy-to-follow responses. They also appreciated the ability of the chatbot to contextualize shared data (from other data sources or chatbot conversations), helping to create a coherent and comprehensible narrative. They also expressed confidence that the chatbot could oversee the sharing of data in a manner that eliminates guessing or misinterpretation. S1 made the following comment: *"Allowing the chatbot to share details gradually is totally different from unloading everything all at once, which often leads to emotional outbursts that spiral out of control. It would be helpful to have a mediator, organizing and summarizing the information and delivering it in a warm and understanding tone. This would be far less stressful and more convenient than having to explain everything on my own."*

3 Preliminary System Design

3.1 Phone Apps

Patient app: As shown in Fig. 1, it features four functional tabs. Home tab (Fig. 1(a)): Clicking on the "Add Test" button allows the patient to self-administer a breathalyzer test [3] and self-report their mood state. This data—as well as lifestyle metrics (e.g., sleep time, activity levels, and screen time) tracked automatically using a wristband tracker [16]—are displayed on a dashboard before being uploaded to a backend server to be shared with a selected family companion. History tab (Fig. 1(b)) provides a weekly summary of self-monitoring data, including alcohol tests, moods, and lifestyle metrics. Patients can navigate through the data to observe patterns and trends in their behavior. The chatbot can be engaged to discuss any data points or relevant topics. Share tab (Fig. 1(c)) organizes logged test results, moods, and lifestyle data into daily cards to facilitate sharing simply by clicking the "Share" button in the bottom-right corner of each card. The chatbot can be engaged to discuss the sharing of specific data entries. Feedback tab (Fig. 1(d)) lists previously shared records, along with summaries generated by the chatbot and responses or encouragement from family companions. Basic chatbot summaries are concise and easy to understand.

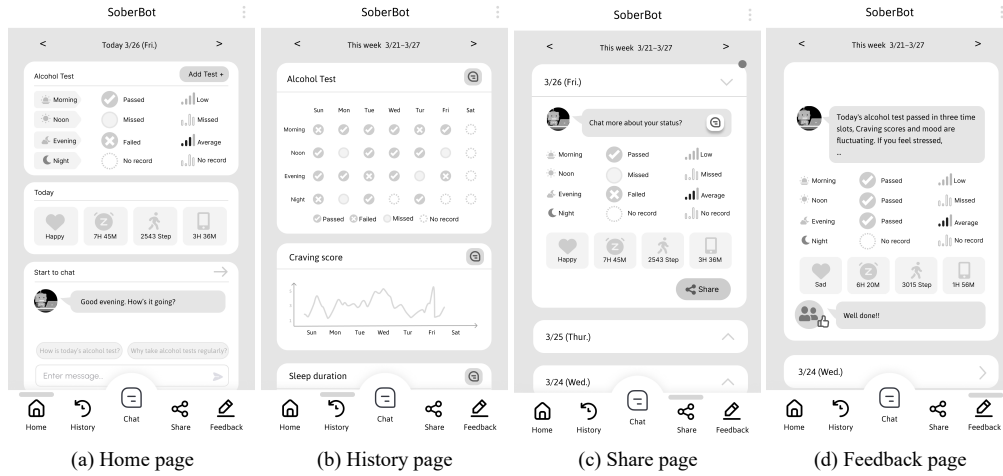


Figure 1: Screenshots of the patient app with its four primary tabs: (a) Home; (b) History; (c) Share; and (d) Feedback.

They can also be linked to daily records with broader contextual factors to present a more comprehensive narrative.

Chat tab (not shown in Fig. 1) is the primary space for engaging with the chatbot with an interface resembling those of popular messaging apps, such as WhatsApp or Facebook Messenger. Patients can address any topic they wish, or click on any data entry in any tab to bring up the chat popup when unsure of how to begin.

Companion app: The app utilizes the chatbot to facilitate family engagement in the recovery process. The app includes four functional tabs. Share tab lists all shared records that have not yet received a response, organized as a list for easy access. Each data entry is grouped within a daily card, which includes alcohol test results, moods, lifestyle data, and a chatbot-generated summary. Clicking the “Details” button opens a detailed summary of the full dataset and provides a space for family companions to leave a response. Feedback tab provides access to previous response records, allowing the tracking of engagements and monitoring of involvement in the recovery process. Each daily data card lists a chatbot-generated summary of the data, followed by visualized details of the daily metrics, and family companion response. This structure facilitates a quick review of relevant information while allowing the chatbot to delve deeper into overlooked details or clarify uncertainties. Chat tab provides direct access to the chatbot to deal with issues related to alcohol use. Clicking on a specific data point initiates a discussion with the chatbot, including a detailed exploration of shared data and corresponding responses. The interface of the Chat tab resembles those used in popular messaging platforms, such as WhatsApp.

3.2 Backend Server and Data Integration

The backend server aggregates data from the patient and companion apps, including alcohol screening results (e.g., pass/fail status and alcohol test results), lifestyle metrics (e.g., sleep hours and activity levels), self-reported emotional states, chat data, and family responses to shared patient data.

The system’s backend incorporates a Retrieval-Augmented Generation (RAG) module connected to a curated knowledge base on Alcohol Use Disorder. This knowledge base—compiled and regularly updated by treatment experts at Taipei City Hospital—includes verified information on AUD fundamentals, coping strategies, relapse prevention, motivational interviewing techniques, and support-group resources. By retrieving the most relevant and up-to-date content at query time, the RAG module ensures that users receive accurate, context-sensitive guidance while minimizing the risk of outdated or misleading information [12].

The server also supports large language model (LLM)-based, context-aware chatbot services. These services deliver personalized support by integrating both real-time and historical user data—including lifestyle metrics, phone usage patterns, self-reported mood, alcohol test results, calendar events, chat history, and time of day. When a user submits a query via the chat popup or Chat tab, the LLM assesses the current conversation context and underlying database schema to determine which structured data could improve its response. It then formulates and executes targeted SQL queries, refines any failed queries automatically, and uses a token-budget manager to select or summarize the retrieved data to fit within the model’s context window. This on-demand pipeline ensures that responses remain focused and responsive to users’ evolving needs.

4 Conclusion

We are currently collaborating with psychiatrists and psychologists to refine the design of the chatbot-assisted, collaborative mobile support system. We will begin recruiting individuals with Alcohol Use Disorder (AUD) and their family companions through referrals from our clinical partners to evaluate the system’s feasibility.

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